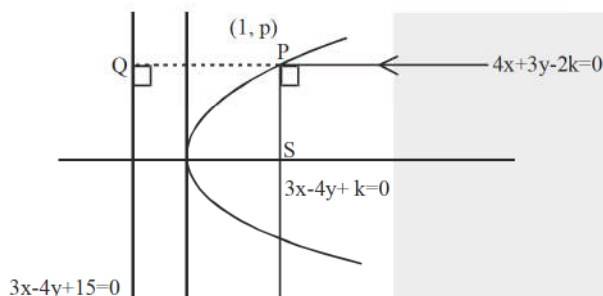


WORK SHEET - 10

PARABOLA

Q.1 4

Sol.



(1, p) satisfies $4x + 3y - 2k = 0$
& $3x - 4y + k = 0$
 $\therefore p = 2$ & $k = 5$
now $P \equiv (1, 2)$.

now $PS = PQ = 2$
 $\therefore LR = 4$

Q.2 3

Sol. We have, $\frac{2}{t_1} \cdot \frac{2}{t_2} = -1 \Rightarrow t_1 t_2 = -4$; also $t_2 = -t_1 - \frac{2}{t_1}$ (1)

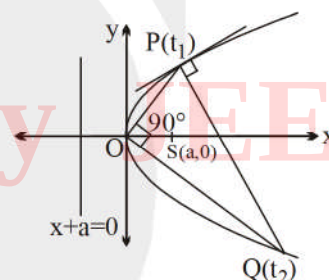
$$t_1 t_2 = -t_1^2 - 2 \Rightarrow -4 + 2 = -t_1^2 \Rightarrow t_1^2 = 2$$

Also, $t_2^2 = t_1^2 + \frac{4}{t_1^2} + 4$ squaring (1)

$$\Rightarrow t_2^2 = 2 + 2 + 4 = 8$$

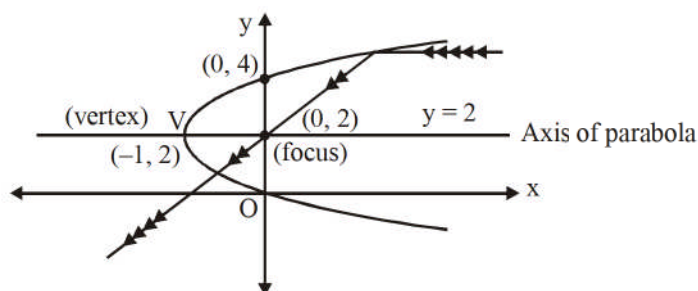
Now, $SQ = a(1 + t_2^2) = a(1 + 8) = 9a$ and $SP = a(1 + t_1^2) = a(1 + 2) = 3a$

$$\therefore \frac{SQ}{SP} = 3 \Rightarrow SQ = 3SP \Rightarrow \lambda = 3 \text{ Ans.}$$



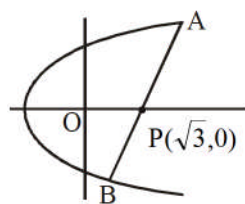
Q.3 (0, 2)

Sol. The equation of the axis of the parabola is $y - 2 = 0$, which is parallel to the x-axis. So, a ray parallel to the x-axis is parallel to the axis of the parabola. We know that any ray parallel to the axis of a parabola passes through the focus of after reflection (using reflection property of parabola). Here, (0, 2) is the focus of parabola.



Q.4 $\frac{4(2+\sqrt{3})}{3}$

Sol. Equation of line can be written as $\frac{x-\sqrt{3}}{\cos \frac{\pi}{3}} = \frac{y-0}{\sin \frac{\pi}{3}} = r$



$\therefore$ Point $= \left(\sqrt{3} + \frac{r}{2}, \frac{r\sqrt{3}}{2} \right)$ lies on parabola.

$$\therefore \frac{3}{4}r^2 = \sqrt{3} + \frac{r}{2} + 2$$

$$\Rightarrow 3r^2 - 2r - 4(2 + \sqrt{3}) = 0$$

$$\therefore PA \cdot PB = r_1 r_2 = \left| \frac{-4(2+\sqrt{3})}{3} \right| = \frac{4(2+\sqrt{3})}{3} \text{ . Ans.}$$

Q.5 $\left(-\frac{2}{3}, 0 \right)$

Q.6 $\frac{\sqrt{5}}{2}$

Sol. Equation of normal at $(t^2, 2t)$ is $(y + tx) = 2t + t^3$
Put $(4, 1)$, we get

$$t^3 - 2t - 1 = 0 \begin{cases} t_1 \\ t_2 \\ t_3 \end{cases}$$

$$\therefore t_1 = \frac{1+\sqrt{5}}{2}, t_2 = \frac{1-\sqrt{5}}{2}, t_3 = -1$$

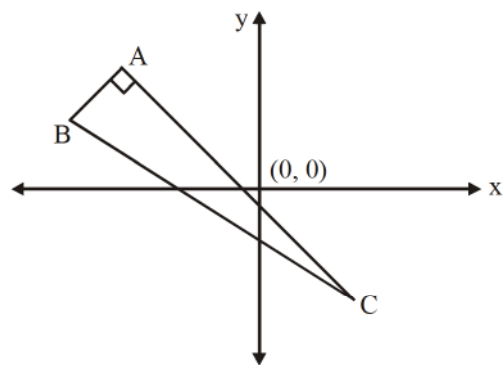
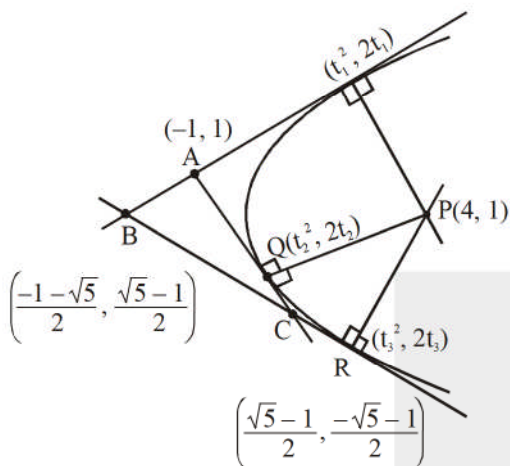
So, $A \equiv (t_1 t_2, t_1 + t_2) \equiv (-1, 1)$

$$B \equiv (t_1 t_3, t_1 + t_3) \equiv \left(\frac{1-\sqrt{5}}{2}, \frac{\sqrt{5}-1}{2} \right)$$

$$C \equiv (t_2 t_3, t_2 + t_3) \equiv \left(\frac{\sqrt{5}-1}{2}, \frac{-\sqrt{5}-1}{2} \right)$$

(i) Centroid of ΔABC ,

$$= \left(\frac{\sum x_i}{3}, \frac{\sum y_i}{3} \right) = \left(-\frac{2}{3}, 0 \right) \text{ Ans.}$$



(ii) $\therefore$ Area of $\triangle ABC$

$$= \frac{1}{2} (AB) (AC) = \frac{1}{2} \left(\frac{\sqrt{20+8\sqrt{5}}}{2} \right) \left(\frac{\sqrt{20-8\sqrt{5}}}{2} \right) = \frac{\sqrt{5}}{2} \quad \text{Ans.}$$

Q.7 $\frac{10}{3}$

Q.8 3

Sol. Tangent to the parabola $y^2 = 4x$

$$y = mx + \frac{1}{m}$$

which is also tangent to the point

$$\therefore p = r$$

$$\left| \frac{-2m + \frac{1}{m}}{\sqrt{1+m^2}} \right| = 1$$

$$\begin{aligned} \Rightarrow |1 - 2m^2| &= m\sqrt{1+m^2} \\ \Rightarrow 4m^4 - 4m^2 + 1 &= m^2(m^2 + 1) \\ \Rightarrow 3m^4 - 5m^2 + 1 &= 0 \\ \Rightarrow m^2 &= \frac{5 \pm \sqrt{25-12}}{6} = \frac{5 \pm \sqrt{13}}{6} \end{aligned}$$

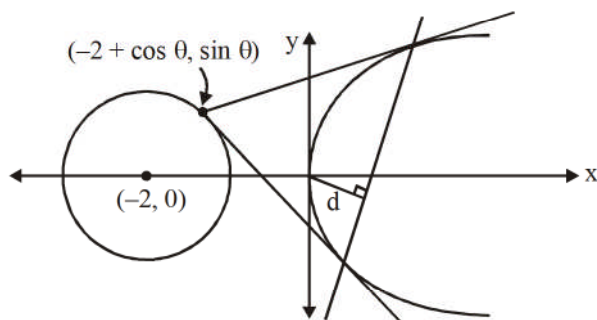
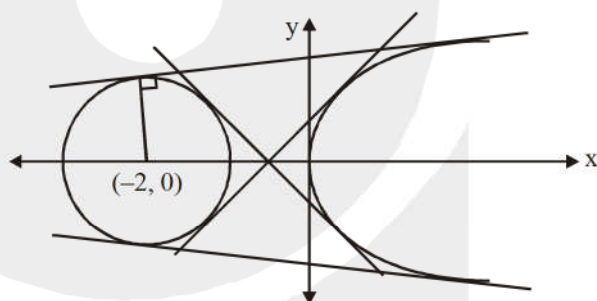
$$\text{Required sum} = \sum_{i=1}^4 m_i^2 = 4 \times \frac{5}{6} = \frac{10}{3}$$

(ii) $(-2 + \cos \theta, \sin \theta)$ is a point on the circle

Chord of contact to the parabola is

$$y \sin \theta = 2 \cdot 1 (x - 2 + \cos \theta)$$

$$2x - y \sin \theta = 2(2 - \cos \theta)$$



∴ Perpendicular distance from the origin is

$$d = \left| \frac{2(2 - \cos \theta)}{\sqrt{4 + \sin^2 \theta}} \right|$$

For the value of d to be maximum, $\theta = \pi$

$$\therefore d|_{\max.} = \frac{2 \times 3}{2} = 3 \text{ Ans.}$$

Q.9 4

Q.10 4

Q.11 $x^2 + y^2 - 8x - 16y = 0$

Sol.

(i) Putting $x^2 = 4y$ in equation of given circle, we get $4y + y^2 - 10y + (25 - r^2) = 0$
 $\Rightarrow y^2 - 6y + (25 - r^2) = 0$ (1)

As parabola is tangent to circle, so discriminant of above equation must be zero.

$$\therefore 36 = 4(25 - r^2) \Rightarrow r = 4 = \text{radius of circle.}$$

(ii) Putting $r = 4$ in equation (1), we get $(y - 3)^2 = 0 \Rightarrow y = 3$.

So, $P(2\sqrt{3}, 3)$ and $Q(-2\sqrt{3}, 3)$.

As directrix of parabola is $x = -1$,

so distance between chord PQ

and directrix $= |3 - (-1)| = 4$. Ans.

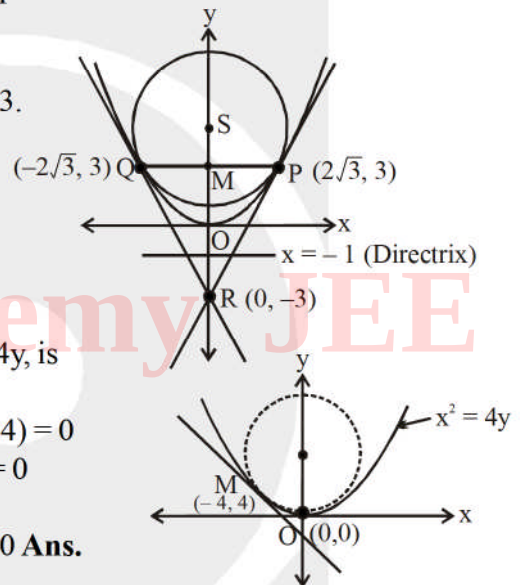
(iii) The equation of tangent at $M(-4, 4)$ to the parabola $x^2 = 4y$, is
 $x(-4) = 2(y + 4) \Rightarrow 2x + y + 4 = 0$

Let the required circle be $(x + 4)^2 + (y - 4)^2 + \lambda(2x + y + 4) = 0$

As above circle passes through $(0, 0)$, so $16 + 16 + \lambda(4) = 0$

$$\Rightarrow 4\lambda = -32 \Rightarrow \lambda = -8$$

Hence required equation of circle is $x^2 + y^2 - 8x - 16y = 0$ Ans.

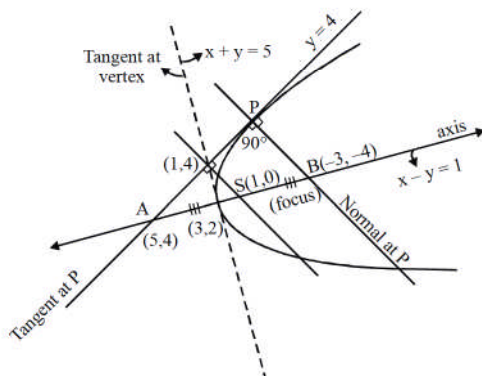


Q.12 $x^2 + y^2 - 2x = 31$

Q.13 $y^2 = 8\sqrt{2}x$

Q.14 64

Sol.



- (i) Tangent and normal are angle bisectors of focal radius and perpendicular to directrix.

$\therefore$ The equation of circle circumscribing ΔAPB , is

$$(x-5)(x+3) + (y-4)(y-4) = 0 \Rightarrow x^2 + y^2 - 2x = 31 \quad \text{Ans.}$$

- (ii) Two parabolas are called equal when their length of latus rectum is same.

Also, $l(L \cdot R) = 4$ (Distance of focus from vertex)

$$= 4\sqrt{(3-1)^2 + (2-0)^2} = 4\sqrt{8} = 8\sqrt{2}$$

- (iii) The area of quadrilateral formed by tangent and normals at ends of latus-rectum $= 8(VS)^2$
 $= 8(4+4) = 8(8) = 64$

Q.15 (A) $x = 0$ (B) $y = \frac{1}{\sqrt{3}}x + \sqrt{3}$ (C) $y = \frac{-1}{\sqrt{3}}x - \sqrt{3}$

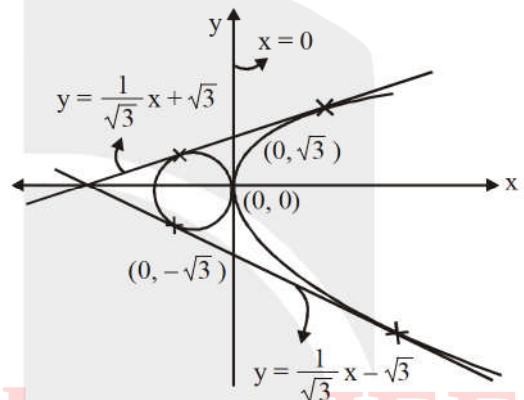
Sol. Any tangent to slope 'm' to $y^2 = 4x$ is

$$y = mx + \frac{1}{m} \text{ or } mx - y + \frac{1}{m} = 0$$

If line (1) is also tangent to circle $(x+1)^2 + y^2 = 1$, then using condition of tangency, we get

$$\begin{aligned} \left(-m + \frac{1}{m}\right)^2 &= (1+m^2) \\ \Rightarrow (1-m^2)^2 &= m^2(1+m^2) \\ \Rightarrow 1-2m^2 &= m^2 \Rightarrow 3m^2 = 1 \\ \therefore m &= \pm \frac{1}{\sqrt{3}} \end{aligned}$$

Also, the line $x = 0$ (called y-axis) is one of their common tangent.



Q.16 (A) Equation of axis of parabola is $x - 4y + 7 = 0$.

(B) Length of Latus rectum of parabola is $\sqrt{17}$.

Sol. Tangents are perpendicular $\Rightarrow AB$ is focal chord

and Normals meet on axis of parabola $\Rightarrow AB$ is double ordinate $\Rightarrow AB$ is latus rectum.

$$\Rightarrow Z(-3, 1)$$

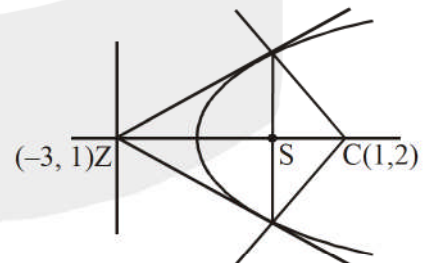
$\therefore$ equation of axis

$$y - 1 = \frac{1}{4}(x + 3)$$

$$4y - 4 = x + 3$$

$$x - 4y + 7 = 0$$

$$CZ = 4a = \sqrt{4^2 + 1^2} = \sqrt{17} \quad \text{Ans.}$$



Q.17 (A) (16, 8) (B) (16, -8)

Sol. Here $a = 1$

$$m = \text{slope of PV} = \frac{2t-0}{t^2-0} = \frac{2}{t}$$

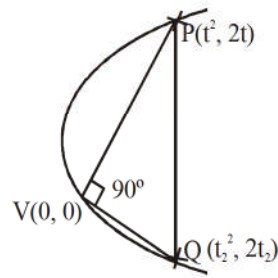


∴ The equation of QV is

$$y = -\frac{t}{2}x \quad \dots (1)$$

∴ On solving (1) with $y^2 = 4x$, we get

$$Q \equiv \left(\frac{16}{t^2}, -\frac{8}{t} \right)$$



$$\text{Now, ar.}(\Delta PVQ) = \frac{1}{2} (PV) (VQ) = 20 \quad (\text{Given})$$

$$\Rightarrow (PV)^2 \times (VQ)^2 = (40)^2$$

$$\Rightarrow \left((t^2)^2 + (2t)^2 \right) \left(\left(\frac{16}{t^2} \right)^2 + \frac{64}{t^2} \right) = 1600$$

$$\Rightarrow (t^2 - 16)(t^2 - 1) = 0$$

$$\Rightarrow t = \pm 4, \pm 1$$

Now, verify alternatives.

Alter

$$\therefore t_1 t_2 = -4$$

$$\text{Area} = \frac{1}{2} \begin{vmatrix} t_1^2 & 2t_1 & 1 \\ t_2^2 & 2t_2 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 20$$

$$\Rightarrow |(t_1 t_2)(t_1 - t_2)| = 20$$

$$\Rightarrow |t_1 - t_2| = 5$$

$$\text{and } t_1 t_2 = -4$$

Solving this $t_1 = \pm 1, t_2 = \pm 4$. Ans.

Q.18 (B) radius of circle described on RS as diameter equals $\frac{7}{2}$.

(C) (1, -1) is a point on the common chord of circle and parabola.

Sol. Given, equation of circle is $(x-2)^2 + y^2 = (\sqrt{10})^2 \quad \dots (1)$

and equation of parabola is

$$y^2 = 8x \quad \dots (2)$$

∴ Put $y^2 = 8x$ in equation (1),

we get $x^2 + 5x - 6 = 0$

$$\Rightarrow x = -6, 1$$

But $x \neq -6$, so $x = 1$

∴ P(1, 3), Q(1, -3)

also, R(-8, 0) and S(-1, 0)

So, the length PQ = 6

Also, radius of circle described on RS as diameter = $\frac{7}{2}$.

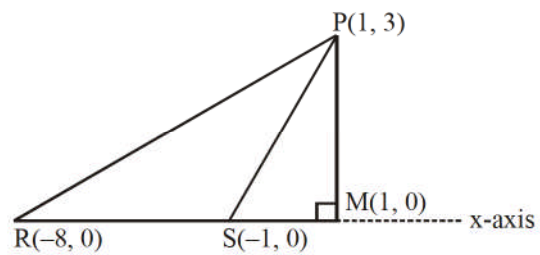
Note that the equation of their common chord

i.e., PQ is the line $x - 1 = 0$.

So, $(1, -1)$ lies on it.

$$\text{So, area } (\Delta PRS) = \frac{1}{2} (RS) (PM)$$

$$= \frac{1}{2} (7) (3) = \frac{21}{2} \quad \text{Ans.}$$



Q.19 2

Sol. Let $y = mx + c$ be the chord

$$\therefore 3x^2 - y^2 - (2x - 4y) \left(\frac{y - mx}{c} \right) = 0$$

$\Rightarrow$ coefficient of x^2 + coefficient of $y^2 = 0$

$$\Rightarrow (y + 2x) + c(x - 1) = 0$$

$\Rightarrow$ fixed point vertex is $(1, -2)$

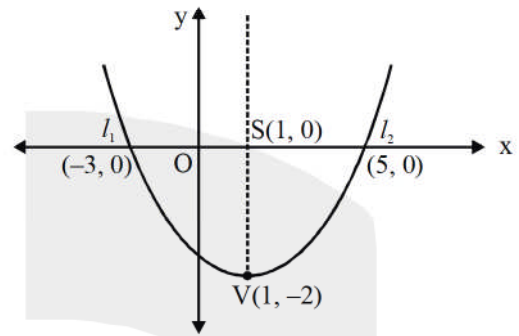
$$\& 4a = 8 = L_1 L_2$$

$$\Rightarrow a = 2$$

$$\therefore L_1 S = L_2 S = 4$$

$$\therefore L_1(-3, 0) \text{ and } L_2(5, 0)$$

$$\therefore \alpha + \beta + \gamma + \delta = 2$$



Q.20 2

Sol. $z = \frac{y-1}{x-\frac{1}{2}} = m_{PQ}$

$$y = mx - \frac{4}{m} \text{ passes } \left(\frac{1}{2}, 1 \right)$$

$$1 = \frac{m}{2} - \frac{4}{m} \Rightarrow m^2 - 2m - 8 = 0$$

$$m^2 - 4m + 2m - 8 = 0$$

$$m = 4, -2$$

$$\Rightarrow M = 4, m = -2$$

$$\therefore |M + m| = 2 \quad \text{Ans.}$$

